# Design regret in clustered wind farms: how wind rose directionality and optimization thoroughness shape the cost of ignoring neighbors

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**Abstract.** When a wind farm is designed in isolation—ignoring the wakes of potential neighboring farms—how much energy is sacrificed? We define this loss as *design regret*: the difference in annual energy production between a layout re-optimized with knowledge of neighbors and the original isolation-designed layout, both evaluated under the same neighbor wakes. Using greedy adversarial neighbor placement and converged multistart optimization ($K = 500$ random restarts), we sweep the two-parameter elliptical wind rose space of Hart (2025) and find that design regret varies six-fold across wind climate: from $17\,\mathrm{GWh/yr}$ ($0.4\,\%$ of AEP) under bidirectional conditions to $101\,\mathrm{GWh/yr}$ ($2.5\,\%$) under concentrated unidirectional conditions. Wind rose directionality—specifically the folding parameter controlling uni- versus bidirectionality—is the dominant driver. Regret cross-sections at $K = 100$ confirm an "ambush effect": peak vulnerability occurs at oblique angles to the dominant wind direction, not along it. Critically, we find that the computational effort required for reliable regret estimation far exceeds standard practice: five random starts capture only $55\,\%$ to (numerical range) $86\,\%$ of the converged regret. We recommend a minimum of 300 multistart optimizations for any study computing AEP differences between design strategies. Buffer distance analysis shows that regret decay depends on wind rose shape: concentrated unidirectional sites require buffers exceeding $40D$ ($9.6\,\mathrm{km}$), while bidirectional sites need minimal separation.

## 1

The North Sea is entering an era of unprecedented wind farm clustering. Denmark's Vindmølleparken energy island will connect up to $10\,\mathrm{GW}$ across multiple farms in a shared concession zone; the German Bight already hosts over $8\,\mathrm{GW}$ with plans for $30\,\mathrm{GW}$ by 2050; and the broader North Sea basin is projected to reach $300\,\mathrm{GW}$ as part of Europe's $500\,\mathrm{GW}$ offshore target. At this density, wind farms no longer operate in isolation. Cluster wake losses in the German Bight are projected to reduce capacity factors by $30\,\%$, with half of the loss attributable to cross-border wake accumulation (Fliegner et al., 2025). The phenomenon—increasingly termed "wind theft" (Finseraas et al., 2024; Du et al., 2026)—has prompted the UK to issue the first national policy guidance on inter-farm wake effects and NYSERDA to recommend minimum inter-farm spacing of $4\,\mathrm{nmi}$ (NYSERDA, 2024). Wake-induced velocity deficits have been measured at distances exceeding $50\,\mathrm{km}$ (Schneemann et al., 2020; Canadillas et al., 2022), and inter-farm losses are projected to exceed the impact of a century of climate change on North Sea wind resources (Warder and Piggott, 2025).

Yet these emerging policy responses share a common gap: they treat all sites uniformly. Buffer distance recommendations, "good neighbour" design principles, and compensation frameworks do not account for variation in wind climate. This is a significant omission. A site where wind blows predominantly from one direction creates a single, predictable wake corridor that a neighbor-aware layout can avoid. A site with bidirectional wind creates competing corridors that no single rearrangement can escape. The value of coordinated design—and the cost of failing to coordinate—should depend fundamentally on the directional structure of the wind resource.

We formalize this intuition through the concept of *design regret*. Following Quick et al. (2026), we distinguish a *liberal* AEP formulation ($\mathrm{AEP}^{\mathrm{lib}}$, internal wakes only) from a *conservative* formulation ($\mathrm{AEP}^{\mathrm{cons}}$, including neighbor wakes). Design regret is the gap in conservative AEP between a layout optimized with neighbor foresight and one optimized in isolation:

$$R = \mathrm{AEP}^{\mathrm{cons}}(\mathbf{x}^*_{\mathrm{cons}}, \mathbf{n}) - \mathrm{AEP}^{\mathrm{cons}}(\mathbf{x}^*_{\mathrm{lib}}, \mathbf{n}) \,. \tag{1}$$

Because both terms are evaluated under $\mathrm{AEP}^{\mathrm{cons}}$, design regret isolates the cost of the design decision itself, separate from the total AEP reduction caused by neighbor wakes. By sweeping the elliptical wind rose parameterization of Hart (2025)—which separates directional concentration from bidirectionality—we quantify how wind rose shape governs design regret. The answer is dramatic: regret varies six-fold across the shape space, from $17\,\mathrm{GWh/yr}$ under bidirectional conditions to $101\,\mathrm{GWh/yr}$ under concentrated unidirectional conditions. Buffer distance requirements inherit this dependence.

Quick et al. (2026) applied this framework to the Danish Energy Island cluster and found negligible design regret for all nine neighbor configurations. We show this is not a general property of offshore wind, but a consequence of DEI's partially bidirectional wind climate ($f \approx 0.38$). Sites in the southern North Sea with strongly unidirectional southwesterly winds sit in the high-regret region of the shape space and are substantially more vulnerable to uncoordinated cluster development.

However, accurately computing these quantities reveals a methodological challenge that undermines all existing regret-like calculations. Layout optimization is highly multimodal (Thomas et al., 2023; Valotta Rodrigues et al., 2024), and regret—as the *difference* between two optimization outcomes—amplifies the effect of local optima. We show that five random starts, more than many published studies employ, capture only $55\,\%$ to (numerical range) $86\,\%$ of the converged regret.

This paper makes four contributions:

1. A convergence study demonstrating that 300+ multistart optimizations are needed for reliable regret estimation—over an order of magnitude more than typical practice.

2. A systematic sweep of the wind rose shape space showing that design regret varies six-fold with directionality, with concentrated unidirectional sites most vulnerable.

3. Buffer distance analysis quantifying how neighbor separation reduces regret, and how the decay rate depends on wind rose shape.

4. Regret cross-sections revealing an "ambush effect" in which peak design vulnerability occurs at oblique angles to the dominant wind direction.

## 2 Methods

### 2.1 Problem formulation

We consider a target wind farm of $N_t$ IEA $15\,\mathrm{MW}$ turbines ($D = 240\,\mathrm{m}$, hub height $150\,\mathrm{m}$) within a fixed polygonal boundary derived from the Danish Energy Island concession zone. Minimum inter-turbine spacing is $4D$ ($960\,\mathrm{m}$). The baseline configuration uses $N_t = 50$; cross-section analyses additionally sweep $N_t \in \{20, 30, 40, 50, 60\}$ to characterize how farm density affects design regret. This configuration is representative of near-future North Sea developments: the IEA $15\,\mathrm{MW}$ turbine is the reference class for planned GW-scale projects, and the DEI polygon provides a realistic, irregular boundary.

The liberal layout $\mathbf{x}^*_{\text{lib}} = \mathrm{argmax}_{\mathbf{x}}\,\text{AEP}^{\text{lib}}(\mathbf{x})$ maximizes AEP assuming no external wakes. The conservative layout $\mathbf{x}^*_{\text{cons}} = \mathrm{argmax}_{\mathbf{x}}\,\text{AEP}^{\text{cons}}(\mathbf{x}, \mathbf{n})$ maximizes AEP including wakes from specified neighbor turbines $\mathbf{n}$. Both layouts are evaluated under $\text{AEP}^{\text{cons}}$ to compute regret via Eq. (1); this holds the physical environment fixed so that only the design decision differs. By construction, $R \geq 0$ when both reach global optima. We enforce this computationally through *regret pooling*: $\text{AEP}^{\text{cons}}(\mathbf{x}^*_{\text{cons}}, \mathbf{n})$ is taken as $\max\big(\text{best conservative AEP},\ \text{AEP}^{\text{cons}}(\mathbf{x}^*_{\text{lib}}, \mathbf{n})\big)$, guaranteeing non-negative regret even with imperfect optimization.

### 2.2 Wake model

We use the Bastankhah and Porté-Agel (2014) Gaussian deficit model with $k = 0.04$ and root-sum-of-squares superposition. The simulation is implemented in JAX (Bradbury et al., 2018), enabling automatic differentiation through the wake model. AEP is computed over 24 equally-spaced wind direction sectors at a constant wind speed of $9\,\mathrm{m/s}$:

$$\text{AEP} = 8760 \sum_{k=1}^{24} w_k \sum_{i=1}^{N_t} P_i(U_{i,k}), \tag{2}$$

where $w_k$ is the sector frequency, $P_i$ is the turbine power curve, and $U_{i,k}$ is the effective wind speed at turbine $i$ under direction $k$. Under $\text{AEP}^{\text{lib}}$, the effective speeds $U_{i,k}$ account only for wakes among the $N_t$ target turbines; under $\text{AEP}^{\text{cons}}$, they additionally include deficit contributions from the neighbor turbines $\mathbf{n}$.

### 2.3 Wind rose parameterization

Wind rose shape is parameterized using the elliptical model of Hart (2025):

- **Shape parameter** $a$: controls directional concentration ($a = 1/\sqrt{\pi} \approx 0.564$ gives uniform).
- **Folding parameter** $f \in [0, 1]$: controls symmetry ($f = 0$ bidirectional; $f = 1$ unidirectional).

The prevailing direction is fixed at $270°$ and wind speed at $9\,\mathrm{m/s}$. We sweep $a \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9\}$ and $f \in \{0.0, 0.25, 0.5, 0.75, 1.0\}$ (35 combinations). This range spans the variety of North Sea wind climates, from the strongly unidirectional southwesterlies of the German Bight to the more bidirectional conditions at the Danish Energy Island.

Fitting the DEI wind rose with the elliptical model yields $a \approx 0.64$, $f \approx 0.38$ ($R^2 = 0.84$), placing it in the moderate-concentration, partially-bidirectional regime.

### 2.4 Adversarial neighbor placement

Neighbors are placed using a greedy grid search. A candidate grid with $5D$ spacing extends $50D$ from the target boundary ($\sim$1567 candidates). At each of 30 greedy steps, the candidate maximizing incremental regret is selected, each requiring a full $K$-start optimization under $\text{AEP}^{\text{cons}}$. This provides an *upper bound* on design regret: it identifies the most vulnerable directions and the maximum cost of designing in isolation, though it does not represent realistic neighbor behavior.

### 2.5 Regret cross-sections

To map directional vulnerability, an identical copy of the target farm—using the liberal-optimized layout ($K = 500$)—is placed at each combination of 24 bearings ($15^\circ$ steps) and 7 buffer distances ($5D$–$60D$), producing 168 evaluations per wind rose. Buffer distance is defined as the minimum Euclidean distance between any point on the target farm boundary polygon and any point on the reference farm boundary polygon; we additionally report the minimum inter-turbine spacing between farms at each configuration. The reference farm thus has the same turbine count, optimized layout, and boundary as the target, differing only in its translated position. At each position, the target farm is re-optimized with $K$ restarts. To characterize how farm density affects vulnerability, we sweep $N_t \in \{20, 30, 40, 50, 60\}$ turbines, re-optimizing the liberal layout with $K = 500$ starts for each farm size. The resulting polar maps—*regret cross-sections*—separate bearing and distance effects and reveal whether vulnerability depends on farm density (Fig. 1).

### 2.6 Multistart optimization and convergence

Each layout optimization uses gradient-based SGD with 5000 iterations (Quick et al., 2023). Restarts include grid initialization, the liberal layout (ensuring regret pooling), and random polygon-sampled positions. Following the convergence analysis of Quick et al. (2026), who found $\sim$400 starts needed for 66-turbine liberal layout convergence, we perform a dedicated convergence study (Section 3.1).

Production configurations: $K = 500$ for the wind rose sweep and buffer analysis; $K = 100$ for cross-sections.

## 3 Results

### 3.1 Convergence study: how many starts are enough?

We sweep $K$ from 1 to 2000 for three configurations spanning the wind rose space (Table 1, Fig. 2). An identical copy of the liberal-optimized target farm is placed at a fixed bearing and buffer distance for each.

At $K = 5$, regret captures only $55\,\%$ to (numerical range) $86\,\%$ of the converged value. The moderate bidirectional case is worst: $K = 5$ finds only $55\,\%$ of true regret because the landscape has many local optima with similar $\text{AEP}^{\text{lib}}$ but different $\text{AEP}^{\text{cons}}$. All configurations stabilize by $K = 300$. SGD iteration convergence is much faster: at $K = 500$, regret at 5000 iterations is within $3\,\%$ of the 10 000-iteration value.

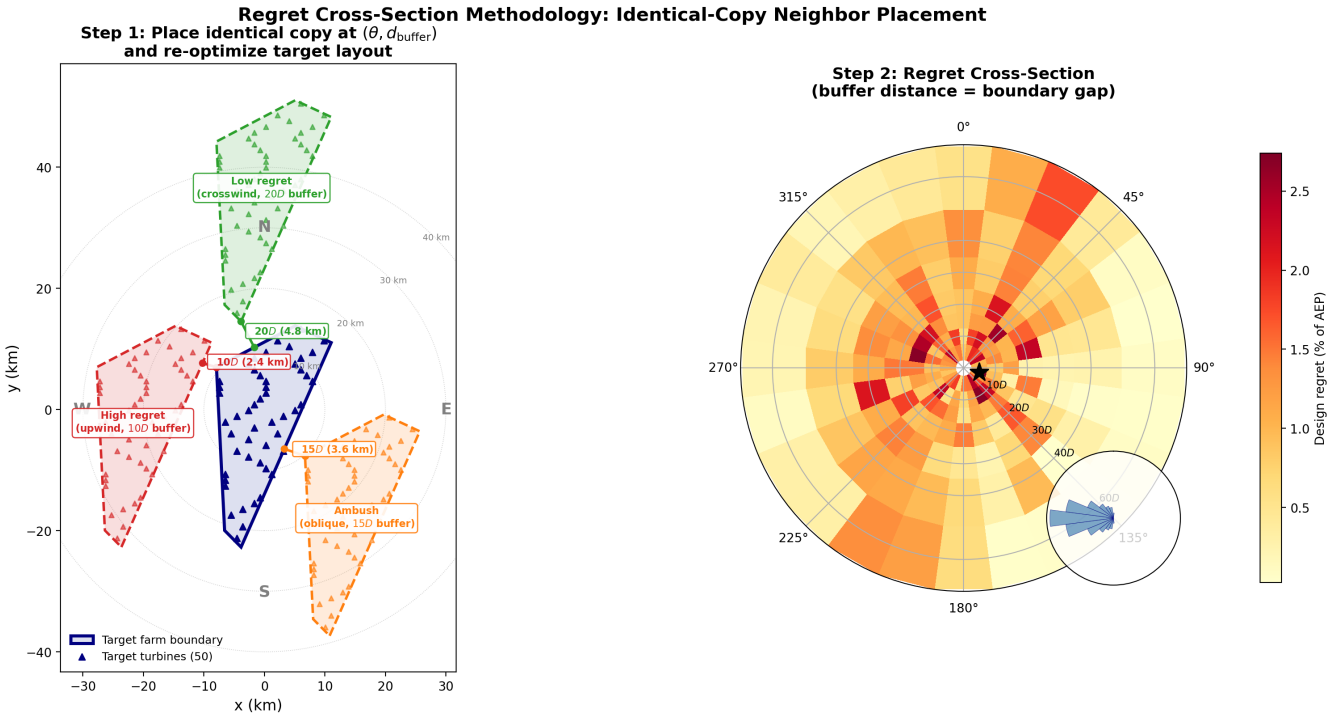


**Figure 1.** Regret cross-section methodology. Left: an identical copy of the liberal-optimized target farm (dashed outlines) is placed at a given bearing and buffer distance from the target (blue polygon), with distance measured between the closest boundary points. Right: the resulting polar map of design regret (% of AEP) as a function of bearing and buffer distance.

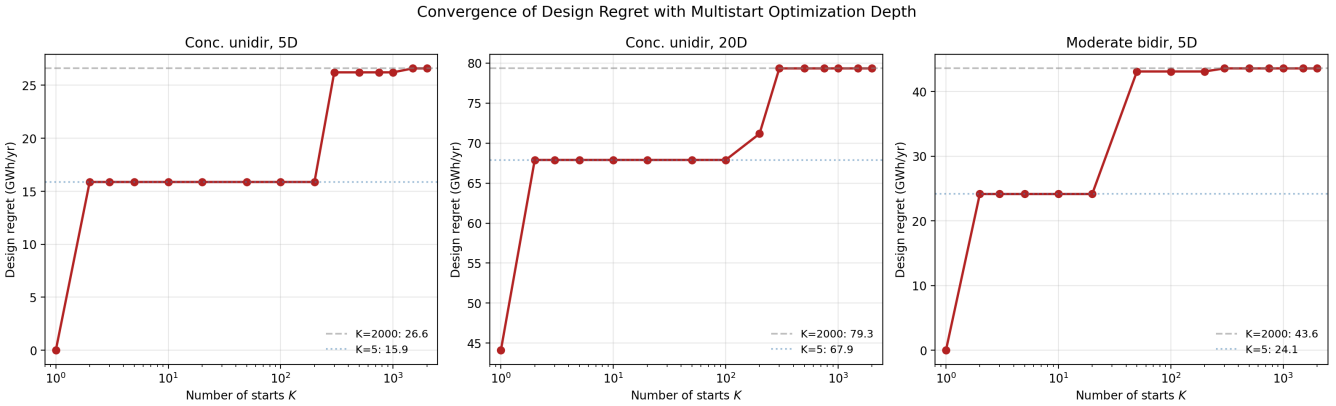


**Figure 2.** Convergence of design regret with multistart depth $K$. Dashed gray: converged value ($K = 2000$). Dotted blue: $K = 5$. All configurations stabilize by $K \approx 300$.

**Table 1.** Design regret (GWh/yr) at selected $K$ values.

| Configuration | $K$=5 | $K$=100 | $K$=300 | $K$=500 | $K$=2000 | $K_{500}/K_{2000}$ |
|---|---|---|---|---|---|---|
| Conc. unidir. ($5D$) | 15.9 | 15.9 | 26.2 | 26.2 | 26.6 | 98% |
| Conc. unidir. ($20D$) | 67.9 | 67.9 | 79.3 | 79.3 | 79.3 | 100% |
| Mod. bidir. ($5D$) | 24.1 | 43.1 | 43.6 | 43.6 | 43.6 | 100% |

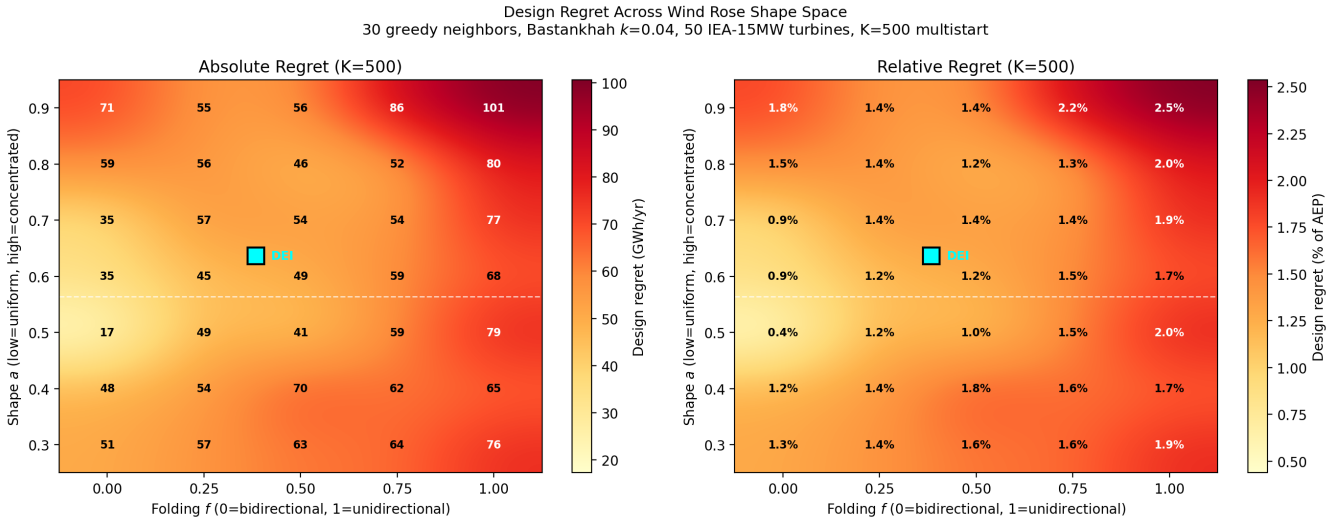


**Figure 3.** Design regret across the wind rose shape space at $K = 500$. Left: absolute (GWh/yr). Right: relative (% of AEP). Cyan square: fitted DEI wind rose ($a \approx 0.64$, $f \approx 0.38$).

The root cause is catastrophic cancellation: regret is a *difference* between two large $\text{AEP}^{\text{cons}}$ values, so a $0.5\,\%$ error in either optimization can produce a $50\,\%$ error in regret. This applies to any study computing AEP differences between strategies—robust optimization, Pareto analysis, or inter-farm coordination assessments.

### 3.2 Wind rose shape controls design regret

Figure 3 shows design regret across the $(a, f)$ space at $K = 500$. Regret ranges from $17.2\,\text{GWh}$ ($0.4\,\%$ of AEP) at $a = 0.5$, $f = 0.0$ to $100.6\,\text{GWh}$ ($2.5\,\%$) at $a = 0.9$, $f = 1.0$—a six-fold range.

The folding parameter $f$ is the dominant factor (Fig. 4). Mean regret increases monotonically from $45.2\,\text{GWh}$ at $f = 0$ to $77.9\,\text{GWh}$ at $f = 1$. Concentration ($a$) has a weaker, non-monotonic effect.

Unidirectional wind ($f = 1$) creates a single wake corridor that the liberal layout is blind to; the conservative optimizer shifts turbines laterally to escape it. Under bidirectional conditions ($f = 0$), wakes arrive from opposing directions and no lateral shift helps—regret is inherently limited.

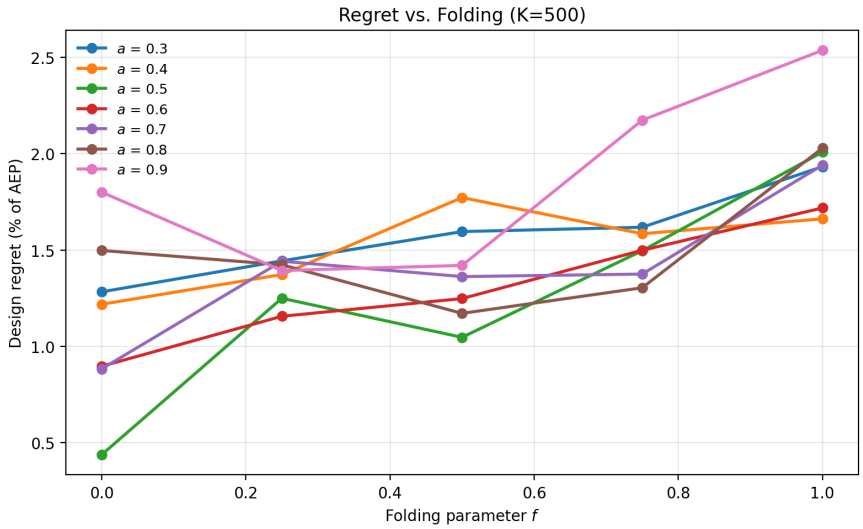


**Figure 4.** Regret vs. folding $f$ at each concentration $a$ ($K = 500$). Monotonic increase with $f$ is consistent across all $a$.

For North Sea planning, this means that sites with strongly unidirectional southwesterly winds (characteristic of the German Bight and southern North Sea) face higher design regret than sites with more bidirectional conditions (such as Danish waters). The DEI wind rose ($a \approx 0.64$, $f \approx 0.38$) falls in the moderate-regret region, consistent with the negligible tradeoffs found by Quick et al. (2026).

### 3.3 Buffer distance modulates regret

Figure 5 shows regret vs. buffer distance at $K = 500$ for three wind roses.

For concentrated unidirectional wind ($a = 0.9$, $f = 1.0$), regret drops from $2.5\,\%$ at $2D$ to $0.4\,\%$ at $40D$ ($9.6\,\mathrm{km}$). The bidirectional case ($a = 0.5$, $f = 0.0$) shows gentle decay from $0.4\,\%$ to $0.2\,\%$.

The NYSERDA $4\,\mathrm{nmi}$ ($\sim 31D$) recommendation reduces regret to $<1\,\%$ for all wind roses tested, but is unnecessarily conservative for bidirectional sites. Wind-rose-dependent buffer standards could allow denser development at low-risk sites (bidirectional, e.g., DEI) while maintaining protection at high-risk sites (unidirectional, e.g., German Bight). This is directly relevant to the emerging cross-border coordination frameworks for the North Sea (Fliegner et al., 2025).

### 3.4 The ambush effect: directional vulnerability

Figure 6 presents regret cross-sections at $K = 100$ for six wind roses (Table 2).

Peak regret consistently occurs at oblique angles to the dominant wind (270°), not along it. We term this the "ambush effect": the liberal layout is pre-adapted to the dominant direction and therefore partially resilient to aligned neighbors. Off-axis neighbors exploit the layout's blind spots—directions the liberal optimizer did not prioritize.

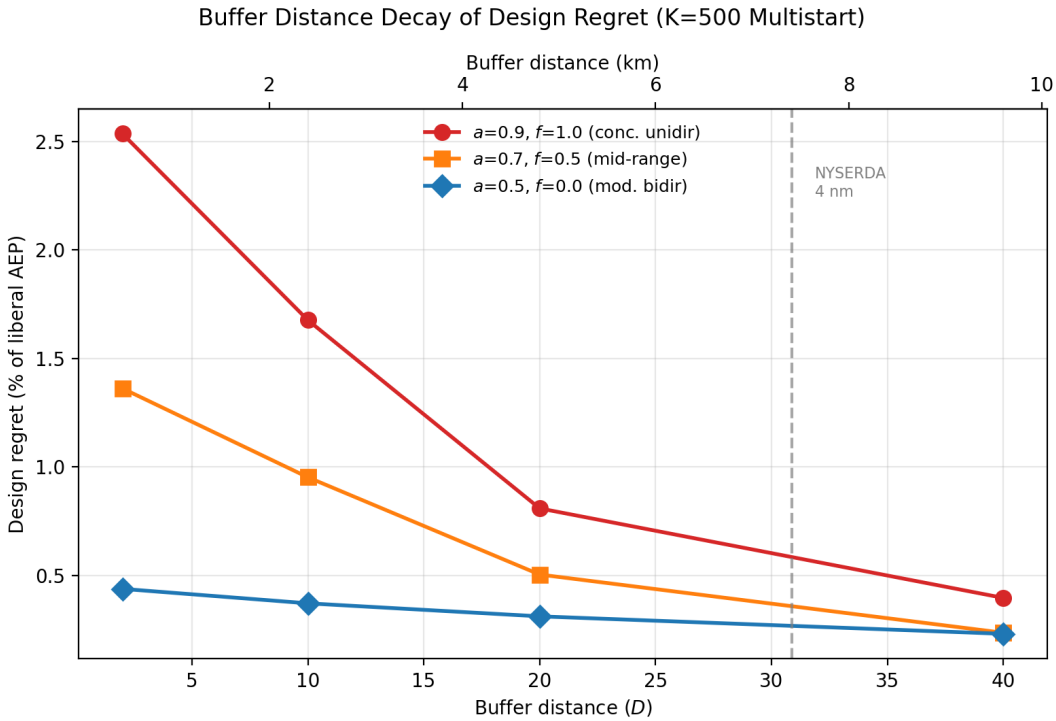


**Figure 5.** Regret vs. buffer distance at $K = 500$, expressed as % of $\mathrm{AEP}^{\mathrm{lib}}$. The NYSERDA 4 nmi recommendation is marked.

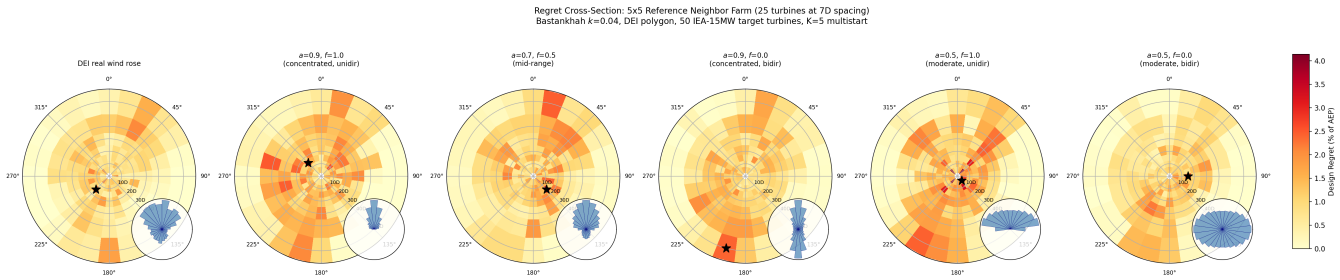


**Figure 6.** Regret cross-sections at $K = 100$: design regret vs. neighbor bearing and buffer distance for an identical-copy reference farm. Stars mark peak regret. Inset wind roses show sector frequencies. Dominant wind: $270^\circ$ (west).

For North Sea cluster planning, this finding is counterintuitive: a developer might assume the most dangerous neighbor is the one directly upwind. The ambush effect shows that neighbors at oblique angles—perhaps in an adjacent country's lease area at a different bearing—may cause greater design regret. This is relevant for cross-border coordination in the North Sea, where multiple national lease areas surround each farm at various bearings.

**Table 2.** Peak regret from cross-sections at $K = 100$. Dominant wind: $270^\circ$.

| Case | Max (GWh) | Max (%) | Bearing | Distance |
|---|---|---|---|---|
| DEI real wind rose | 122.3 | 2.21 | $225^\circ$ | $15D$ |
| $a = 0.9$, $f = 1.0$ | 117.6 | 2.96 | $315^\circ$ | $15D$ |
| $a = 0.7$, $f = 0.5$ | 114.2 | 2.89 | $135^\circ$ | $15D$ |
| $a = 0.9$, $f = 0.0$ | 95.0 | 2.39 | $195^\circ$ | $60D$ |
| $a = 0.5$, $f = 1.0$ | 163.3 | 4.14 | $135^\circ$ | $5D$ |
| $a = 0.5$, $f = 0.0$ | 91.2 | 2.31 | $90^\circ$ | $15D$ |

These cross-sections use $K = 100$, which the convergence study shows may underestimate converged regret by up to $40\,\%$ for some configurations. The qualitative directional pattern is robust—it is physically grounded and consistent across all six cases—but absolute magnitudes should be interpreted as lower bounds.

# 4 Discussion

## 4.1 Implications for North Sea cluster development

The six-fold variation in design regret across wind rose shape has direct implications for the North Sea's emerging cluster framework.

### 4.1.1 German Bight and southern North Sea.

Sites with strongly unidirectional southwesterly winds sit in the high-regret region ($f \gtrsim 0.75$). Here, the value of coordinated design is highest, and buffer distances of $30$–$40D$ ($7\,\mathrm{km}$ to (numerical range) $10\,\mathrm{km}$) may be needed to reduce regret below $1\,\%$. The projected $30\,\%$ capacity factor reduction from cluster wakes (Fliegner et al., 2025) is partially a design problem, not just a physics problem: some of this loss could be mitigated by neighbor-aware layout optimization.

### 4.1.2 Danish waters.

The DEI wind rose ($f \approx 0.38$) benefits from a natural resilience: wakes from any single direction are partially offset by energy captured from the opposing direction, limiting the rearrangement advantage of a neighbor-aware design. This explains the negligible design tradeoffs found by Quick et al. (2026). However, this resilience is a property of the wind climate, not the farm—other Danish sites with different wind characteristics may not share it.

#### 4.1.3 Cross-border coordination.

The value of inter-farm coordination is itself wind-rose-dependent. Our results suggest that coordination frameworks like the "good neighbour" principle in the UK EN-3 guidance should be weighted by wind rose directionality. Fixed buffer distances—such as the NYSERDA $4\,\mathrm{nmi}$ recommendation—are adequate for all sites tested but inefficient: bidirectional sites need only $\sim 10D$ to achieve the same residual regret. Wind-rose-conditional buffers could allow denser development at low-risk sites without compromising high-risk ones.

### 4.2 The convergence problem

Our most broadly applicable finding is methodological. At $K = 5$, regret estimates can be off by nearly a factor of two. The cause is catastrophic cancellation: regret is a difference between two large, noisy $\mathrm{AEP}^{\mathrm{cons}}$ values. A $0.5\,\%$ miss in either optimization can produce a $50\,\%$ error in the difference.

This affects any study computing AEP differences between strategies. We recommend $K \geq 300$ for 50-turbine farms and proportionally more for larger farms. The computational cost at $K = 500$ was approximately $2\,\mathrm{hours}$ per wind rose on a single AMD MI250X GCD, or $70\,\mathrm{GPU-hours}$ for the full 35-case sweep.

### 4.3 Limitations

#### 4.3.1 Wake model.

We use Bastankhah and Porté-Agel (2014) with $k = 0.04$ and root-sum-of-squares superposition. More sophisticated models (e.g., TurbOPark) may change absolute magnitudes. The qualitative findings—folding dominance, the ambush effect—depend on wake corridor geometry, not magnitude, and are expected to be robust.

#### 4.3.2 Adversarial assumption.

The greedy grid search provides an upper bound. Realistic self-interested neighbors may produce different regret patterns; preliminary evidence suggests they can produce *higher* regret at certain bearings because AEP-optimal neighbor layouts concentrate turbines upwind, but this requires further investigation at converged $K$.

#### 4.3.3 Constant wind speed.

Real North Sea sites have speed–direction correlations. If high-speed winds are concentrated in the dominant direction (common for storm-driven sites), regret could be amplified through the cubic power relationship.

#### 4.3.4 Single farm geometry.

Results may depend on the DEI polygon shape. Different boundaries could affect the layout optimization landscape and the potential for lateral escape from wake corridors.

#### 4.3.5 Cross-section $K$ level.

Cross-sections use $K = 100$, which may underestimate converged regret. Directional patterns are robust; magnitudes are lower bounds.

## 5

1. **Standard multistart practice drastically underestimates design regret.** Five random starts capture only $55\,\%$ to (numerical range) $86\,\%$ of converged regret. All configurations stabilize by $K = 300$, with no change through $K = 2000$. Any study computing AEP differences between strategies should use $K \geq 300$.

2. **Design regret varies six-fold across the wind rose shape space.** At $K = 500$, regret ranges from $17\,\mathrm{GWh/yr}$ ($0.4\,\%$) under bidirectional conditions to $101\,\mathrm{GWh/yr}$ ($2.5\,\%$) under concentrated unidirectional conditions. Folding ($f$, bidirectional vs. unidirectional) is the primary driver.

3. **Buffer distance requirements are wind-rose-dependent.** Concentrated unidirectional sites retain $0.4\,\%$ residual regret at $40D$ ($9.6\,\mathrm{km}$). Bidirectional sites achieve comparable levels at $10D$ ($2.4\,\mathrm{km}$). Wind-rose-conditional buffer standards could allow denser North Sea development at low-risk sites.

4. **Peak vulnerability occurs at oblique angles to the dominant wind.** The "ambush effect" means that off-axis neighbors—perhaps from adjacent countries' lease areas—may cause greater design regret than directly upwind neighbors. This is relevant for cross-border coordination in the North Sea.

The Danish Energy Island, with its partially bidirectional wind rose ($f \approx 0.38$), sits in the low-to-moderate regret region. Sites in the southern North Sea with strongly unidirectional southwesterly winds are substantially more vulnerable. As North Sea clustering intensifies toward the $300\,\mathrm{GW}$ target, incorporating wind rose directionality into buffer distance standards and coordination frameworks will become increasingly important for efficient maritime spatial planning.

. The simulation framework (pixwake) and analysis scripts are available at https://github.com/kilojoules/cluster-tradeoffs. The elliptical wind rose parameterization uses the edrose package (Hart, 2025), available at https://github.com/kilojoules/edrose.

. JQ designed the study, developed the simulation framework, performed all computations, and wrote the manuscript. PER supervised the research and contributed to methodology and manuscript revision.

. The authors declare no competing interests.

. Computational resources were provided by the LUMI supercomputer, hosted by CSC – IT Center for Science (Finland) and the LUMI consortium.

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